

Section 22: VLAN & Inter-VLAN Routing

Layer: L2/L3 | Reference: Odom Vol1 Ch12-14: VLANs, VLAN Trunking, InterVLAN Routing | *Original study notes — CCNA 200-301*

VLANs let you segment a physical switch into multiple logical broadcast domains without needing separate physical hardware for each department or function. Routing between VLANs requires Layer 3 intervention since VLANs are isolated broadcast domains by design.

Key Concepts

- A VLAN is a logically separate broadcast domain configured on a switch; devices in different VLANs cannot communicate without a Layer 3 device routing between them
- VLAN 1 is the default VLAN on every Cisco switch and carries untagged traffic by default - best practice is to avoid using VLAN 1 for production traffic for security reasons
- Access ports carry traffic for exactly one VLAN and are typically used for end devices (PCs, printers, etc.)
- Trunk ports carry traffic for multiple VLANs simultaneously, tagging frames with 802.1Q headers so the receiving switch knows which VLAN each frame belongs to
- The native VLAN on a trunk carries untagged traffic; native VLAN mismatches between two ends of a trunk cause VLAN leakage and are a classic troubleshooting scenario
- Router-on-a-Stick (ROAS): a single router interface is divided into sub-interfaces, one per VLAN, each with 802.1Q encapsulation matching a VLAN tag - used to route between VLANs with only one physical link to the router
- SVI (Switch Virtual Interface): a virtual Layer 3 interface configured directly on a multilayer switch for a VLAN, eliminating the need for an external router for inter-VLAN routing
- DTP (Dynamic Trunking Protocol) automatically negotiates trunk status between two connected switches - generally recommended to disable in production and hard-code trunk/access mode for predictability and security
- VTP (VLAN Trunking Protocol) synchronizes VLAN database information across multiple switches in a domain, but a misconfigured VTP can accidentally wipe VLANs across an entire domain - use with caution

Command Reference

Command	Purpose
<code>vlan <id> / name <name></code>	Create a VLAN and assign it a name (Global Config / VLAN sub-mode)
<code>switchport mode access</code>	Set a port to access mode (single VLAN, untagged)
<code>switchport access vlan <id></code>	Assign an access port to a specific VLAN
<code>switchport mode trunk</code>	Set a port to trunk mode (carries multiple tagged VLANs)
<code>switchport trunk native vlan <id></code>	Set the native (untagged) VLAN on a trunk

Command	Purpose
switchport trunk allowed vlan <list>	Restrict which VLANs are permitted across a trunk
interface <intf>.<subif> / encapsulation dot1q <id>	Create a router sub-interface for ROAS, tagged for a specific VLAN
interface vlan <id> / ip address <ip> <mask>	Create an SVI on a multilayer switch for inter-VLAN routing
show vlan brief	List all VLANs and which ports are assigned to each
show interfaces trunk	Verify trunk status, native VLAN, and allowed VLAN list

Configuration Walkthrough

Create VLANs and assign access ports

```
SW1(config)#vlan 10
SW1(config-vlan)#name SALES
SW1(config)#interface FastEthernet0/5
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
```

Router-on-a-Stick: sub-interfaces for two VLANs on one physical link

```
R1(config)#interface GigabitEthernet0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config)#interface GigabitEthernet0/0.20
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
```

Inter-VLAN routing via SVI on a multilayer switch

```
SW1(config)#ip routing
SW1(config)#interface vlan 10
SW1(config-if)#ip address 192.168.10.1 255.255.255.0
SW1(config-if)#no shutdown
```

Worked Scenario

Q: Two switches are trunked together, but hosts in VLAN 99 on SW1 can unexpectedly see broadcast traffic intended for VLAN 1 on SW2. What's the likely misconfiguration?

A native VLAN mismatch on the trunk. If SW1's trunk has native VLAN 99 configured while SW2's trunk still uses the default native VLAN 1, untagged frames sent on each side get interpreted as belonging to the OTHER switch's native VLAN once received - causing VLAN leakage between what should be isolated broadcast domains. Fix by setting matching native VLAN values on both ends of every trunk.

Common Mistakes / Gotchas

- Forgetting that VLANs are isolated broadcast domains by design - a router or SVI is mandatory for any inter-VLAN communication
- Leaving native VLAN mismatched across a trunk, causing VLAN leakage
- Not realizing 'ip routing' must be explicitly enabled on a multilayer switch before SVIs will actually route between VLANs
- Using VLAN 1 for sensitive production traffic despite it being the universal default/native VLAN target for many attacks

Exam Tips

- ★ *ROAS (sub-interfaces + router) vs SVI (built into the switch) is a direct architecture comparison the exam tests*
- ★ *Know that trunk ports tag frames with 802.1Q, except for the native VLAN, which stays untagged*
- ★ *Practice configuring both ROAS and SVI from memory - they're common simlet tasks*